

☐ Salesforce Summer Program – Day 10 (4–5 Hours)

☐ Goal for Today

Today is the MOST IMPORTANT transition day.

Now you will integrate:

- CRM
- Data Modeling
- Validation Rules
- Flows
- Apex
- SOQL
- Triggers
- LWC

into ONE connected mini project.

☐ Deep Learning Modules (IMPORTANT)

1 ☐ Build a Simple LWC Application

Focus:

- Connecting UI with backend
 - Real application structure
 - Data flow
 - User interaction
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2 ☐ Lightning Web Components and Salesforce Data

Focus:

- Data retrieval
 - User interaction
 - Reactive UI
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☐ **Light Completion Modules**

3☐ **Visualforce Basics (Overview)**

4☐ **Aura Components Basics (Overview)**

☐ **Videos (Verified Working Links)**

1☐ **LWC Full Course for Beginners**

<https://www.youtube.com/watch?v=IHezETiVWPo>

Focus:

- LWC architecture
 - Component structure
 - Real examples
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2☐ **Salesforce LWC Project Tutorial**

<https://www.youtube.com/watch?v=SygSlOIT7fU>

Focus:

- Real project thinking
 - UI + backend integration
 - Data flow
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3☐ **LWC Communication Explained**

<https://www.youtube.com/watch?v=2Kk2m0wKk9E>

Focus:

- Parent-child communication
 - Events
 - Component interaction
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4 ☐ Aura vs LWC Explained

<https://www.youtube.com/watch?v=9m4r2xZlK0U>

Focus:

- Why Salesforce modernized UI architecture
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☐ CORE MINI PROJECT (MANDATORY)

☐ College Management Mini Project

Integrate ALL concepts learned till now.

REQUIRED FEATURES

CRM Concepts

- Student
 - Faculty
 - Course
 - Department
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Data Modeling

- Objects
 - Relationships
 - Fields
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Validation Rules

Examples:

- Email mandatory
- Seats cannot exceed limit

Formula Fields

Examples:

- Remaining seats
- Attendance percentage

Flow Automation

Examples:

- Auto confirmation email
- Attendance warning

Apex Logic

Examples:

- Eligibility calculation
- Bulk operations

LWC UI

Design:

- Student Dashboard
- Faculty Dashboard
- Registration Screen

Trigger/Event Thinking

Examples:

- Notify faculty when course full
 - Alert for low attendance
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❑ SYSTEM THINKING TASKS

Task 1 ❑ Complete Data Flow

Explain:

Student clicks Register →

LWC screen →

Validation →

Flow →

Trigger →

Database →

Notification

❑ Explain complete flow clearly.

Task 2 ❑ Architecture Thinking

Why do enterprise systems need:

- frontend
 - backend
 - database
 - automation
 - events
 - all together?
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Task 3 ❑ Scaling Thinking

Suppose:

- 50,000 students use system

What problems might happen?

Examples:

- Performance
 - Data consistency
 - Notifications
 - Security
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Task 4 Reflection Task

What did you realize about enterprise software systems after learning Salesforce?

GitHub Submission

Create:

/day10-mini-project/

README.md must include:

- System Overview
 - CRM Concepts
 - Data Model
 - Validation Rules
 - Flows
 - Apex Logic
 - UI Screens
 - Complete Data Flow
 - Reflection
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Revision Questions

1. Why do enterprise systems need modular architecture?
 2. Why are relationships important?
 3. Why are Flows insufficient for some cases?
 4. Why do systems need event-driven behavior?
 5. Why is UI/backend separation important?
 6. Why do enterprise systems require testing?
 7. Why is reusable UI architecture powerful?
 8. What problems happen when systems scale?
 9. Why should automation be designed carefully?
 10. How do all Salesforce concepts integrate together?
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End of Day Outcome

You should now clearly understand:

- ☐ Enterprise application architecture
 - ☐ UI/backend/data integration
 - ☐ Automation + events + business logic
 - ☐ Modular application design
 - ☐ How real Salesforce systems are structured
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